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SC4016: Cyber Threat Intelligence

Group 3

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Abstract

Our assignment involved conducting an investigation into various ransomware shaming groups and extracting information from the shaming websites to gain insights into the data they have exfiltrated and to address several pertinent enquiries.

Ransomware Group	Count	Website
Arcus Media	28	arcuufpr5xxbbkin4mlidt7itmr6znlppk63jbtkeguuhszmc5g7qdyd[.]onion
BianLian	130	bianlianlbc5an4kgnay3opdemgcryg2kpfcbgczopmm3dnbz3uaunad[.]onion/
Hunter	109	hunters55atbdusuladzv7vzv6a423bkh6ksl2uftwrxyuarbzlfh7yd[.]onion/
Play	201	k7kg3jqxang3wh7hnmaiokchk7qoebupfgoik6rha6mjpzwupwtj25yd.onion/
QiLin	20	kbsqoivihgdmwczmxkbovk7ss2dcynitwhhfu5yw725dboqo5kthfaad[.]onion/
TOTAL	**488**	

Table.1 - Ransomware group

1. What is the % distribution in countries targeted globally?

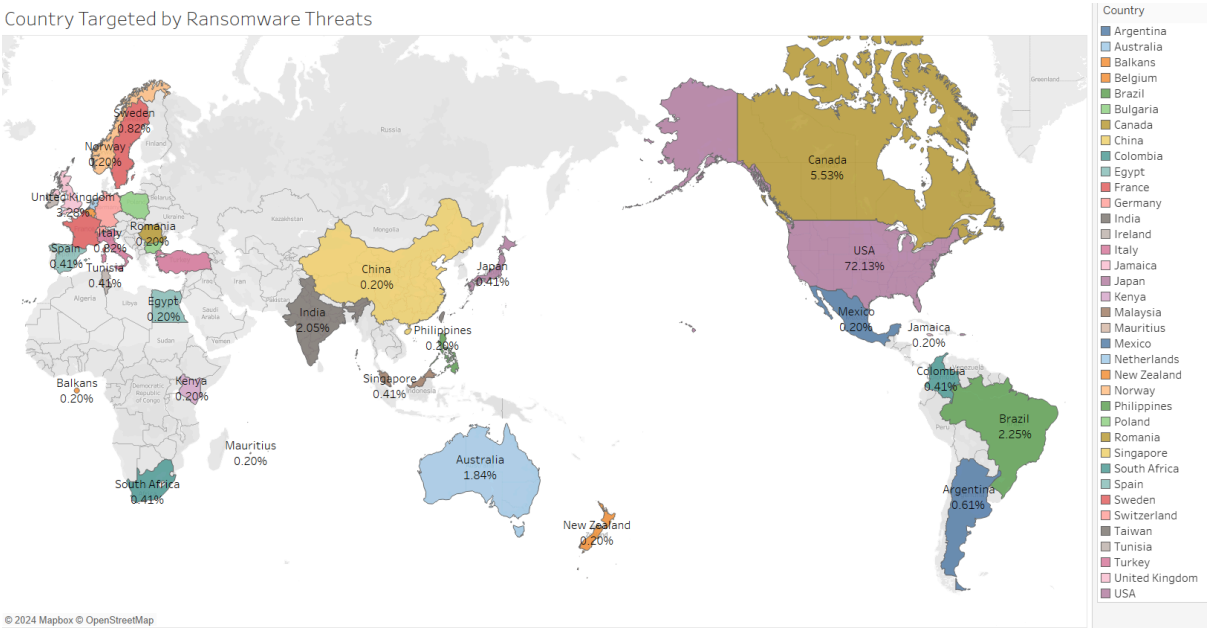

Fig.1 - Countries affected by Ransomware Group

We can observe the percentage of each country targeted from a total of 37 Countries affected.

1. USA: 72.13%
2. Canada: 5.58%
3. United Kingdom: 3.28%

2. Why are some countries more targeted than others?

Hypothesis & Evidence can be found in Appendix A.

Assessment

Evidence	H1: Larger Economic Impact	H2: Higher Digital Dependence	H3: Political Motivation or Retaliation	H4: Weaker or Inconsistent Cybersecurity Across Industries	H5: Higher Reporting Standards and Transparency	H6: More Lucrative Insurance Payouts and Willingness to Pay Ransoms
E1: U.S has largest economy	++	+	-	--	+	--
E2: High internet and cloud adoption	NA	++	--	--	--	--

E3: State-sponsored actors targeting the U.S.	++	++	++	--	--	--
E4: SMEs lacking cybersecurity investment	-	+	+	++	--	++
E5: High adoption of cyber insurance	--	-	--	--	--	++
E6: U.S possess valuable data	++	-	++	--	-	+

Refinement and Inconsistency can be found in Appendix B.

Conclusion

The United States of America (USA) was the focus because it comprised most attacks (72.13%) in a country targeted by ransomware.

After considering the above Evidence and Hypothesis, we concluded that the USA provided a Larger Economic Impact (H1) and having a likelihood of Political Motivation or Retaliation (H3) was the primary reason why they were attacked the most with probable confidence. The USA possesses a substantial quantity of relevant data at its disposal (E6). Either their military data or the intellectual property (IP) of American companies constitutes vital assets for other nations to advance their technology and economy. This was one of the important pieces of evidence that was taken into consideration. Additional evidence suggested the potential involvement of state-sponsored actors in targeting the United States (E3).

3. Which ransomware group is the most active? What is so unique about their TTP that makes them so "successful"?

Play ransomware is known for targeting unpatched vulnerabilities, such as those found in Microsoft Exchange servers. This includes exploiting critical vulnerabilities like CVE-2022-41082, which grants attackers access to servers by bypassing security patches. The group is particularly opportunistic in using unpatched systems to gain entry into networks, a strategy that allows them to compromise organisations quickly (CISA, 2023)

Tactics	Techniques	Elaboration
Initial Access	[T1078] Abuse of valid accounts [T1190] Exploitation of public facing applications	PLAY abuses valid accounts on victim's networks and exploits public-facing applications such as FortiOS and

	(CVE-2018-13379 (FortiOS), CVE-2020-12812 (FortiOS), CVE-2022-41040 (Microsoft Exchange) and CVE-2022-41082(Microsoft Exchange)) [T1133] External Remote services (RDP, VPN)	Microsoft Exchange vulnerabilities. External-facing services such as RDP and VPN are also used.
Discovery	[TA0007] Active Directory Queries (Discovery) [T1016] System Network Configuration Discovery [T1518.001] Scan for antivirus software (software Discovery)	Ransomware actors also attempt to gain more information about the victim's network infrastructure and anti-virus software through various scans.
Defence Evasion	[T1562.001] Impair defences (through GMER, IOBit, and PowerTool) [T1070.001] Clear logs (windows event logs)	Anti-virus software is then disabled and log files are removed to keep their tracks hidden.
Credential Access	[T1152] Unsecured Credentials [T1003] OS Credential Dumping (Mimikatz Credential Dumper)	Threat actors search for unsecured credentials and use credential dumpers to obtain privileged access.
Lateral Movement & C2	[T1570] Lateral Tool Transfer (distribute executables via group policy objects) [T1484.001] Domain Policy Modification (Group Policy)	Subsequently, ransomware executables are proliferated using Group Policy Objects.
Collection	[T1560.001] Archive Collected data (Winrar to compress file)	Upon obtaining the information, ransomware actors compress the data through Winrar, potentially bypassing the size limit of files that can be exfiltrated.
Exfiltration	[T1048] Exfiltration Over Alternative Protocol WinSCP to transfer file	WinSCP is then employed to exfiltrate the compressed data.
Impact	[T1486] Data Encrypted [T1657] Financial Theft	Data is encrypted with RSA-AES hybrid encryption and double extortion is carried out to compel victims to pay.

Table.2 - MITRE ATT&CK Framework mapping for PLAY (CISA, 2023)

PLAY ransomware's successes can be attributed to its many different anti-analysing techniques, making reverse engineering of these malwares an arduous task. Anti-sandboxing functions that cause timeouts are written into the ransomware to prevent dynamic analysis of these binaries in sandboxes. Garbage codes are also written into the program to obfuscate the control flow of the ransomware and names of APIs used are also hashed, hampering static analysis. Additionally, its encryption technique is also notably unique and never before seen. For instance, its encryption and decryption keys are never stored within the malware itself and are instead generated dynamically during runtime through a complex series of nested functions (Novick, 2024). The above factors ensure that PLAY's ransomware is practically unbreakable while leaving little room for researchers to analyse its inner workings

4. Which industries are more prone to ransomware threats? Why?

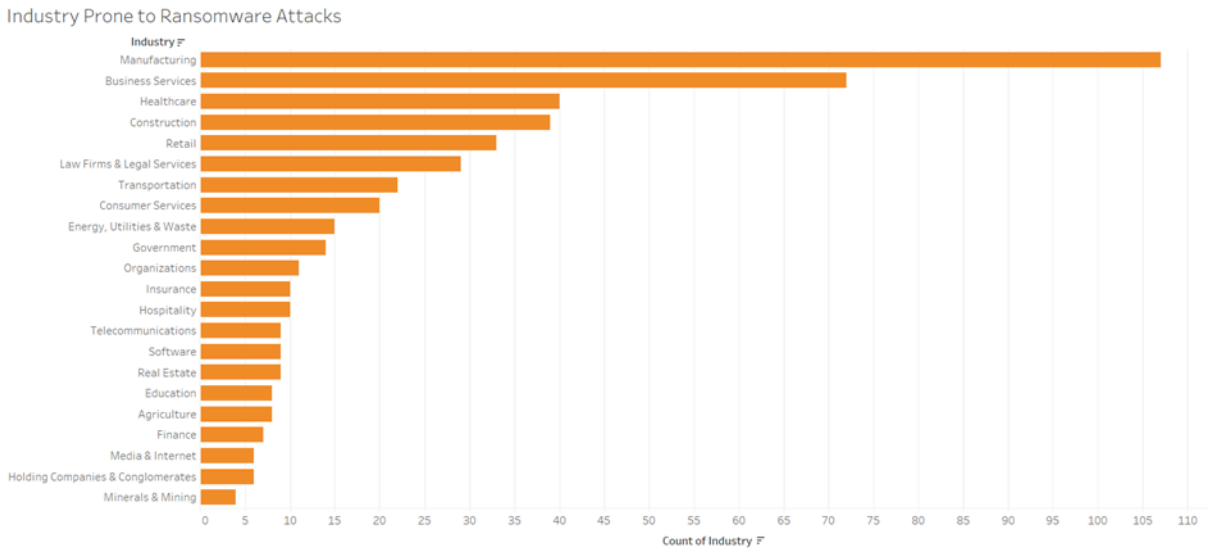

Fig.2 - Industries affected by Ransomware Group

The industries that are more prone to ransomware threats are Manufacturing, Business Services, and Healthcare. In manufacturing, outdated legacy systems (David, 2024), complex interconnected networks, and the proliferation of IoT devices create expansive attack surfaces (Richards, 2024), leading to significant revenue disruptions during downtime. Business services firms are targeted for their sensitive client data, making successful attacks not only damaging to the firm but also to its customers while eroding trust and reputation (Mason, 2024). Meanwhile, healthcare services struggle with limited cybersecurity resources and outdated systems, complicating updates and increasing susceptibility (Wolff, 2024). The interconnected nature of hospital systems means that ransomware attacks can disrupt patient care, highlighting the critical need for enhanced cybersecurity measures across these industries (Olsen, 2024).

Hypothesis

H1: <u>Usage of legacy systems</u>

Industries such as manufacturing and healthcare are limited to the use of legacy systems because of software and device interconnectivity

H2: <u>Willing to pay the ransom</u>

These industries cannot afford to have their systems locked out or shut down as it can lead to disruption of services or production. For the healthcare industry, patient lives are at stake.

H3: <u>Political motivation or retaliation</u>

Certain ransomware groups may be state-sponsored or politically motivated, targeting industries to disrupt their economy or critical services.

H4: <u>Weaker cybersecurity amongst industries</u>

Healthcare services have limited resources and expertise to devote to cybersecurity. Other than that, small businesses do not have a devoted cybersecurity team as cybersecurity is not a primary concern for them.

H5: <u>Attack surface expansion</u>

Manufacturing and healthcare industries use a lot of interconnected systems leading to an increase in attack surface

Evidence

Label	Factor (Reports)	Explanation (Summary of Reports)
E1	Outdated systems in use	Research has shown that companies rely on older legacy systems for operations, often reluctant to upgrade to newer software systems due to cost, potential downtime, and compatibility issues. Healthcare institutions are often forced to run software that is compatible with older equipment and systems that they rely on for patient care. Trying to update systems causes compatibility issues with equipment, forcing healthcare institutions to use legacy systems. Prevalent in many manufacturing infrastructures often lack robust security protocols and are therefore susceptible to ransomware attacks. 74% of manufacturing and engineering companies post-pandemic are relying on legacy systems and spreadsheets to get tasks done (The Manufacturer, 2022).
E2	Large attack surface	Hospitals operate in an increasingly internet-connected environment, connecting various healthcare systems leading to a “target-rich” attack surface Manufacturing facilities are frequently built on a complex network of interconnected systems, machinery, inventory delivery, and data repositories, providing a “target-rich” and expansive attack surface for ransomware perpetrators.
E3	Massive financial losses	66% of organisations reported a significant loss of revenue following a ransomware attack. 35% of businesses that paid a ransom demand shelled out between $350,000 to $1.4 million, while 7% paid ransoms exceeding $1.4 million. According to a study by Cybereason, the average cost of a ransomware attack on a manufacturing company is approximately $1.85 million, excluding the potential reputational damage and loss of consumer trust (David, 2024). According to a study by IBM Cost of a Data Breach Report, Healthcare has the most expensive data breach recovery around $9.77 million on average (Bonderud, 2024).
E4	Insufficient cybersecurity efforts	SMBs (Small to Medium Businesses) are often underfunded in cybersecurity efforts. Reports show that many organisations lack proper defence systems and employee training against phishing and ransomware. 60% of small businesses do not have a proper cybersecurity policy. Hospitals have limited resources and expertise to devote to cybersecurity. Healthcare staff aren’t educated on online risks. Medical professionals do not have the necessary expertise to recognize and mitigate basic threats. The X-Force report reveals that in attacks on critical infrastructure organisations, 85% of the incidents could have been mitigated with patching, multi-factor authentication, or least privilege principles (Alvarez, 2024).
E5	Motives against industry	Some ransomware groups, like those linked to Russia, may have indirect support from state actors, increasing attacks on politically adversarial nations like the U.S. This can lead to motivation to attack major industries to disrupt society on a large scale, apart from financial losses.

Assessment

Evidence	H1: Usage of legacy systems	H2: Willing to pay the ransom	H3: Political motivation or retaliation	H4: Weaker cybersecurity amongst industries	H5: Attack surface expansion
E1: Outdated systems in use	++	++	NA	++	++
E2: Large attack surface	+	+	-	+	++
E3: Massive financial losses	++	++	-	++	++
E4: Insufficient cybersecurity efforts	++	+	NA	++	+
E5: Motives against industry	NA	+	++	+	-

Refinement and Inconsistency can be found in Appendix B.

Conclusion

From the above hypothesis, we concluded with high confidence that the Manufacturing, Business Services, and Healthcare industries are more prone to ransomware threats. The usage of legacy systems (H1), and willingness to pay the ransom (H2) are considered. Many companies in these three industries have outdated systems in use (E1), which enables attackers to target them easily. Because they cannot afford to have their systems locked out or shut down which can lead to the disruption of services and production, they often are willing to pay the ransom (H2) and this leads to massive financial losses (E3).

NOTE: Questions 5 & 6 used data from only BianLian and PLAY because the other Ransomware sites did not provide relevant data necessary to perform further analysis.

5. Actors target sensitive data, but what kind of data do actors usually target? What kinds of data are targeted at each industry? Show a breakdown comparing types of data stolen.

Types of Data	Confidential Business Information (CBI)
	Financial Information
	Intellectual Property (IP)
	Personally Identifiable Information (PII)

Industries	Business Services
	Construction
	Consumer Services
	Energy, Utilities & Waste
	Healthcare
	Insurance
	Law Firms & Legal Services
	Manufacturing
	Retail
	Transportation

Table.3 - Data Types and Industries

What kinds of data do threat actors target for each industry?

- Based on the information that we have found, threat actors generally tend to steal a wide variety of data and not just a specific type from a specific company. For example, Battle Lumber Co (Construction Industry), had CBI, PII, and financial data stolen - Finance data, HR data, Partners and vendors data, Clients and customers data, Engineering, PII and PHI records, Incidents and accidents data, Mailboxes, and internal & external email correspondence.
- Shaming sites seem to prioritise the volume of data rather than the quality of data. This could be because shaming sites do not directly use the data extracted, rather the data is meant to be used by other parties or to flex the capabilities of the hacker group. Thus, they are less concerned with streamlining or organising the data extracted.
- However, some data types tend to be more popular among different industries than others. CBI, Financial, and PII data are widely stolen in every industry, while IP data is least stolen by the companies. This is also understandable as not many companies have their own product which is patented.

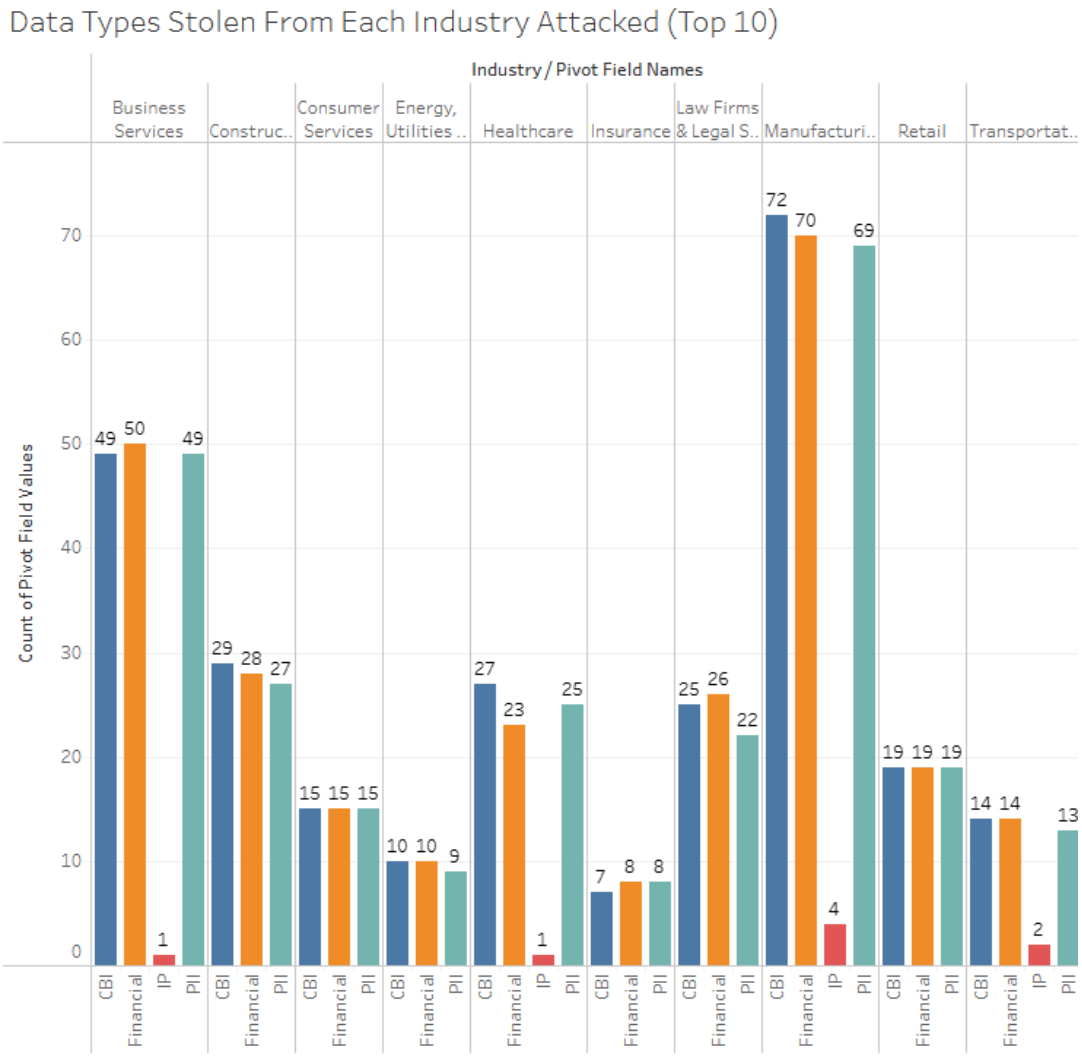

Fig.3 - Data types stolen by industry

6. Share 3 interesting insights you observed

1. From previous questions, it is observed that a majority of BianLian's victims are from the USA. Given that the name coined for the ransomware attack group is similar to Chinese characters (变脸), could there perhaps be another ulterior motive for attacking victims located in the USA besides financial gains.

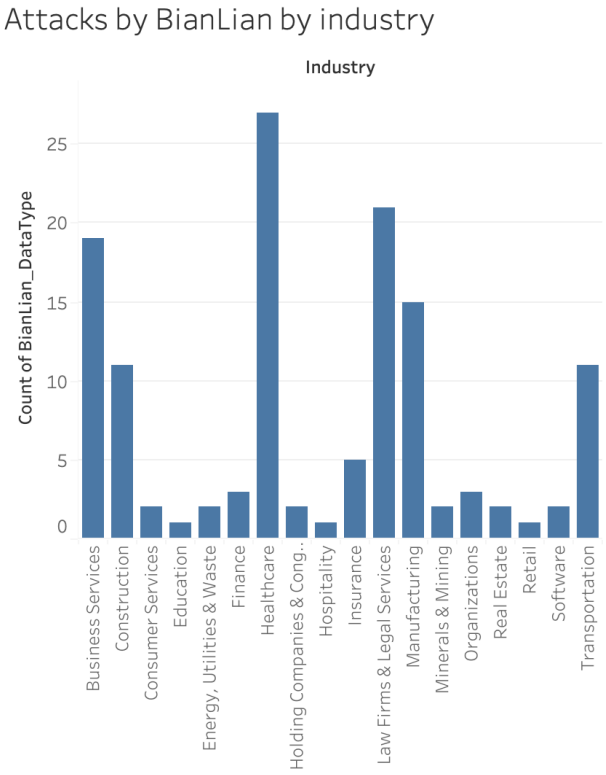

Fig.4 - BianLian attacks by Industry

Based on our findings, healthcare industries were attacked the most at 27 counts. BianLian repeatedly attacked critical public health care infrastructures such as the Texas Centers for Infectious Disease Associates and Center for Digestive Health, which can potentially point to political motives. However, it was found that the group communicated in Russian and none of the victims were ever from a Russian-speaking country, contradicting the above hypothesis (Barry, 2024). Based on the information gathered, it is hard to confidently attribute BianLian to any country. For now, it may seem like they are just a ransomware group looking for financial gains, but the possibility of BianLian being a state-sponsored threat actor cannot be ruled out with absolute certainty.

2. Ransomware actors try to get whatever sensitive information they can get a hold of to 'persuade victims to pay money' (Xcitium, 2024). Attackers seem to dump all the information they have access to, barring having to compromise their operation security (OP Sec).
   2.1. Most companies have more than 1 type of data stolen.

2.2. Threat actors seem to extract data indiscriminately, with some hackers extracting Terabytes of information for a specific company

2.3. Interestingly, no Credentials and IT system information were published in all the cases of ransomware attacks we gathered. However, this does not mean that the threat actors do not have a hold of such data. This data may be unpublished to maintain the attackers' OP Sec if the victims let their guard down and not update security measures or change credentials. This is especially so given that ransomware attacks on the same victims by the same group happen quite frequently (Coker, 2024).

2.4. A disproportionate amount of IP data was stolen compared to other data types. As shown previously in question Q5, the manufacturing industry suffered the most ransomware attacks. Theoretically, IP data theft should be substantial as most manufacturing companies have some form of IP data which includes product designs and materials (Zhurer, 2022). Additionally, IP data is often substantially more valuable than other digital data in the company (WIPO, n.d.), causing companies to take additional precautions in protecting said data. For example, companies may avoid storing IP digitally entirely, given the prevalence of ransomware attacks and data theft from digital infrastructures.

3. 42% of victim companies are small companies, and fall victim to ransomware attacks due to the fact that with fewer employees and clients, security is not an immediate priority to these companies, causing them to often opt to reduce spending on security. 60% of small businesses do not have a cybersecurity policy and therefore are in potential danger of being attacked at any point in time. Ransomware attackers can exploit such small businesses knowing that the business owners would pay the ransom (Davies, 2024).

   Small businesses can also allow attackers to target larger companies using a supply chain attack or watering hole attack. By exploiting the trusted partners of larger companies, threat actors can gain a foothold in the networks of larger companies without having to bypass their cyber defence capabilities directly. Attackers could also use smaller companies to farm credentials to the networks to larger companies (Pourkhomami, 2021). A famous example of this is the 2013 cyber attack on Target, where credentials were stolen from a third party vendor instead of Target itself (Joyce, 2017).

   This phenomenon extends to medium-sized companies as well, as the number of attacks targeting each type of victim seems inversely proportional to their size. This seems to confirm the intuitive hypothesis that larger companies have more resources to build cybersecurity capabilities. Despite being more enticing targets for shaming sites, they seem to be far harder for hackers to exploit (Pourkhomami, 2021).

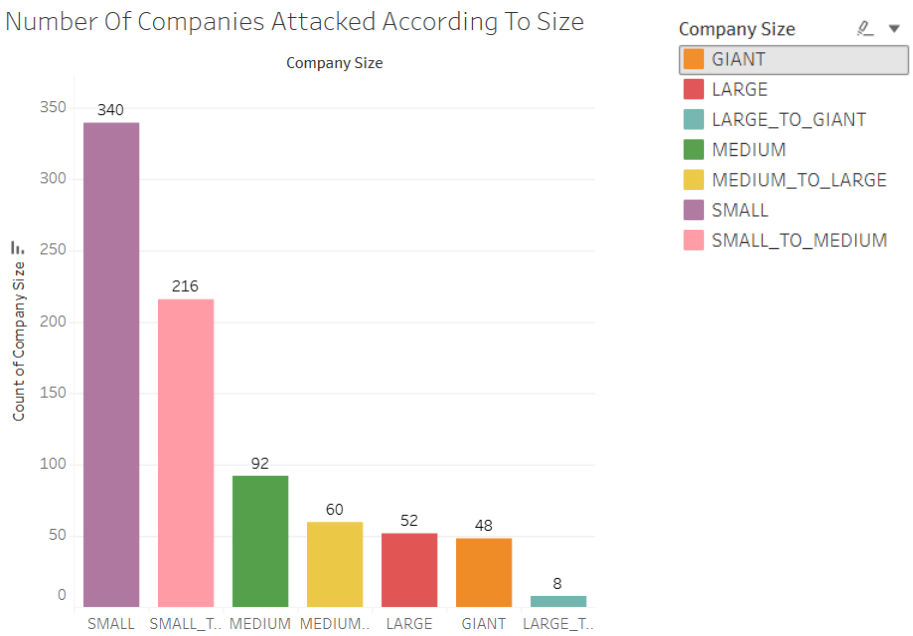

Fig.5 - Company size vs Companies attacks

7. Share lessons learnt, what were your struggles in executing the project and how did you overcome them?

1. Shaming websites can be difficult to scrape due to their varying formats and frameworks.

   Hunters website in particular was difficult to scrape due to the website using the Angular framework. This resulted in the webpage only rendering an HTML page mid-runtime rather than passing it as a simple HTTP payload. To overcome this challenge, our scraper for Hunter had to utilise the selenium utility to render the web page and automate interaction with the website to ensure that all web page elements are properly rendered before scraping.

2. Missing information and cleaning data we scrapped. After scraping the different shaming sites, we used Glassdoor API to find information on the companies that were attacked. Expectedly, the Glassdoor API was not able to provide full information for each of the companies, such as revenue, company size, industry, etc. We had to take extra measures to fill in the missing information as much as possible before we could analyse the data in Tableau. There was still some unfilled information, and our visualisations on tableau may not be fully representative of the true information.

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Appendix A

Hypothesis

H1: Larger Economic Impact

The U.S. is targeted more frequently because it has the largest economy and the potential financial gains are higher for attackers.

H2: Higher Digital Dependence

The U.S. is highly digitised, with many organisations relying on digital infrastructure, making them more susceptible to ransomware attacks.

H3: Political Motivation or Retaliation

Certain ransomware groups may be state-sponsored or politically motivated, targeting the U.S. to disrupt its economy or critical services (Ribeiro, 2024).

H4: Weaker or Inconsistent Cybersecurity Across Industries

Despite overall stronger cybersecurity in the U.S., there are weak spots, particularly in small to medium-sized businesses (SMBs) that make up a large part of the economy and may not be well-protected.

H5: Higher Reporting Standards and Transparency

U.S. companies are legally required to report data breaches and ransomware incidents, leading to more visibility of attacks compared to other countries where incidents may go unreported.

H6: More Lucrative Insurance Payouts and Willingness to Pay Ransoms

Companies in the U.S. are more likely to have cyber insurance and, consequently, might be more willing to pay ransoms, making them attractive targets for attackers.

Evidence

Label	Factor (Reports)	Explanation (Summary of Reports)
E1	U.S. has largest economy	The U.S. has a GDP over $26 trillion, and ransomware attackers often aim to extort high-value organisations like Fortune 500 companies. Attacks on companies like Colonial Pipeline show the high potential for financial gains (Rasure, 2024).
E2	High internet and cloud adoption	The U.S. leads globally in internet usage and cloud adoption. Critical infrastructure (healthcare, energy, finance) in the U.S. is heavily digitised, creating many attack surfaces.
E3	State-sponsored actors targeting the U.S.	Some ransomware groups, like those linked to Russia, may have indirect support from state actors, increasing attacks on politically adversarial nations like the U.S. Sanctions and trade tensions between the U.S. and certain nations could be a motivating factor behind the attacks.
E4	SMEs lacking cybersecurity investment	SMBs in the U.S. are often underfunded in cybersecurity efforts. Reports show that many of these organisations lack proper defence systems and employee training against phishing and ransomware (Office Of The National Cyber Director Executive Office Of The President, 2024).
E5	High adoption of cyber insurance	Cyber insurance is widely adopted in the U.S. Studies indicate that companies that are insured are more likely to pay ransoms to restore their data, especially when insurance covers part of the cost.

E6	US possess valuable data	The US is a global superpower, and hence is an extremely important player in geopolitics. This significance elevates the value of military data of the US, as it is of interest to many other countries. China for example, is frequently charged with computer intrusions targeting US (Office of Public Affairs, 2024). The China-US rivalry also extends to US based companies as they also hold personal data of US citizens (TUCKER & BALSAMO, 2020). US also has the 2nd highest number of patent applications with 17.2% (WIPO, 2024). Furthermore, 38.2% of the US GDP comes from intellectual property-intensive industries. As a result, Intellectual property theft costs the US $225 billion to $600 billion annually (CRI Group, 2024) . Also, 8 out of the Top 10 largest companies by Market Cap are from the USA, implying that they hold the most personal data internationally (Companies Market Cap, 2024).

Refinement and Inconsistency

Evidence	**H1: Larger Economic Impact**	**H2: Higher Digital Dependence**	**H3: Political Motivation or Retaliation**
E1: U.S has largest economy	**++**	**+**	**-**
E2: High internet and cloud adoption	**NA**	**++**	**--**
E3: State-sponsored actors targeting the U.S.	**++**	**++**	**++**
E4: SMEs lacking cybersecurity investment	**-**	**+**	**+**
E5: High adoption of cyber insurance	**--**	**-**	**--**
E6: U.S possess valuable data	**++**	**-**	**++**

Appendix B

Refinement and Inconsistency

Evidence	**H1: Usage of legacy systems**	**H2: Willing to pay the ransom**	**H4: Weaker cybersecurity amongst industries**	**H5: Attack surface expansion**
E1: Outdated systems in use	**++**	**++**	**++**	**++**
E2: Large attack surface	**+**	**+**	**+**	**++**
E3: Massive financial losses	**++**	**++**	**++**	**++**
E4: Insufficient cybersecurity efforts	**++**	**+**	**++**	**+**